

Unit 4 :: Special Theory of Relativity

☞ **PHY0400104: Classical Mechanics** ☞

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Syllabus

Unit 4: Special theory of relativity

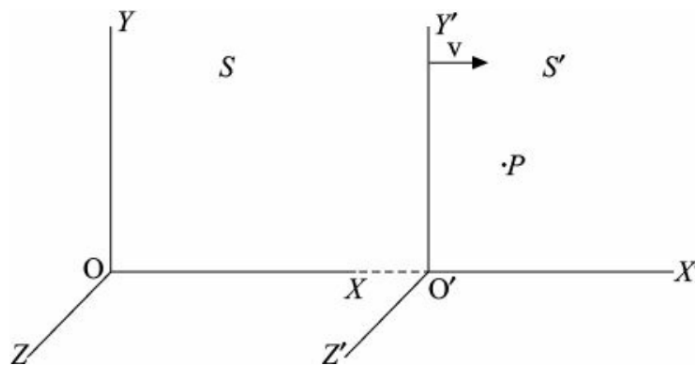
Inadequacy of Galilean transformation; postulates of special relativity; Lorentz transformation; simultaneity and order of events; length contraction and time dilation; relativistic addition of velocities; variation of mass with velocity and mass-energy equivalence. Lorentz transformation as a rotation in spacetime; relation between proper time and coordinate time; relativistic kinematics: energy-momentum relation.

1 Galilean transformation

An *inertial frame* is one in which the law of inertia holds good.

- An ideal inertial frame is a coordinate frame of reference which is fixed in space with respect to the fixed stars.
- A set of coordinate axes attached to the Earth may be regarded as an *almost* inertial frame of reference.

An accelerating frame of reference is known as a *non-inertial frame*.



Consider two inertial frames S and S' with coordinate axes $x y z$ and $x' y' z'$ attached to them, with respective origins O and O' respectively. Let the frame S' be moving with respect to S with a uniform velocity v along the $x x'$ axes as shown in the figure above. The origins of the two frames are assumed to coincide when $t = t' = 0$.

Let an event be taking place at P whose coordinate with respect to S be (x, y, z, t) , and with respect to S' be (x', y', z', t') . Then these coordinates are related by the following *Galilean transformation equations*:

$$\begin{cases} x' = x - v t \\ y' = y \\ z' = z \\ t' = t \end{cases} \quad (1)$$

With the assumption that $\frac{d}{dt} = \frac{d}{dt'}$, we differentiate equation (1) with respect to time to obtain the *Galilean velocity transformations*:

$$\begin{aligned} \frac{dx'}{dt'} &= \frac{d}{dt}(x - v t) \\ \frac{dy'}{dt'} &= \frac{dy}{dt} \\ \frac{dz'}{dt'} &= \frac{dz}{dt} \end{aligned} \quad \Rightarrow \quad \begin{cases} u'_x = u_x - v \\ u'_y = u_y \\ u'_z = u_z \end{cases} \quad (2)$$

Again we differentiate equation (2) with respect to time to obtain the *Galilean acceleration transformations*:

$$\begin{aligned} \frac{d^2 x'}{dt'^2} &= \frac{d^2 x}{dt^2} \\ \frac{d^2 y'}{dt'^2} &= \frac{d^2 y}{dt^2} \\ \frac{d^2 z'}{dt'^2} &= \frac{d^2 z}{dt^2} \end{aligned} \implies \boxed{\begin{aligned} a'_x &= a_x \\ a'_y &= a_y \\ a'_z &= a_z \end{aligned}} \quad (3)$$

Because the directions of the unit vectors are parallel in both frames, we have $\hat{\mathbf{x}}' = \hat{\mathbf{x}}$, $\hat{\mathbf{y}}' = \hat{\mathbf{y}}$ and $\hat{\mathbf{z}}' = \hat{\mathbf{z}}$. Then from equation (3), we find that

$$\begin{aligned} a'_x \hat{\mathbf{x}}' + a'_y \hat{\mathbf{y}}' + a'_z \hat{\mathbf{z}}' &= a_x \hat{\mathbf{x}} + a_y \hat{\mathbf{y}} + a_z \hat{\mathbf{z}} \\ \implies \mathbf{a}' &= \mathbf{a} \end{aligned} \quad (4)$$

i.e. the acceleration of a particle remains invariant under a Galilean transformation.

Multiplying equation (4) by the mass m of the particle, we have

$$\begin{aligned} m\mathbf{a}' &= m\mathbf{a} \\ \implies \mathbf{F}' &= \mathbf{F} \end{aligned} \quad (5)$$

Equation (5) implies that the force on a body remains invariant across inertial frames of reference.

Principle of Galilean-Newtonian relativity: No inertial frame is special and all inertial frames are equivalent.

1.1 The inadequacy of Galilean transformation

As per Maxwell's electromagnetic theory, an electromagnetic (EM) wave propagates through free space with a constant speed of $c \approx 3 \times 10^8$ m/s.

Then the wavefront of an EM wave emitted by a point source at O would be spherical with a radius at time t to be ct . Hence the spherical wavefront surface is given in the frame S to be

$$\begin{aligned} x^2 + y^2 + z^2 &= (ct)^2 \\ \implies x^2 + y^2 + z^2 - c^2 t^2 &= 0 \end{aligned} \quad (6)$$

For the Galilean-Newtonian principle to hold, we must have that the above equation is invariant between S and S' , and so we must have

$$x'^2 + y'^2 + z'^2 - c^2 t'^2 = 0 \quad (7)$$

Substituting equations (1) in equation (7), we get

$$(x - vt)^2 + y^2 + z^2 - c^2 t^2 = 0 \quad (8)$$

But equation (8) is not identical to equation (6). This inconsistency seems to suggest that there must be some special reference frame with respect to which the speed of light is 3×10^8 m/s.

In order to support this suggestion, it was assumed that light and other EM waves are mechanical waves, which thereby require a medium to propagate. This medium was given the name *ether*, was assumed to have the following properties:

- i. Transparent and massless
- ii. Permeates all space
- iii. Rigid enough to support transverse motion of EM waves

The ether hypothesis led to the following two alternatives:

1. *The stationary ether hypothesis*: where the ether is at rest with respect to the bodies moving through it; the speed of light is always c in the ether frame.
2. *The ether drag hypothesis*: where the ether is dragged along with the bodies that move through it.

A number of experiments were performed to verify the ether hypothesis and all had failed. The most direct experiment was the Michelson-Morley experiment.

2 The postulates of special relativity

Due to the failure to detect the ether medium, Albert Einstein dropped the concept of ether, along with the assumption of an absolute rest frame.

Einstein revised the classical ideas of space and time in the form of the following two *postulates of special relativity*:

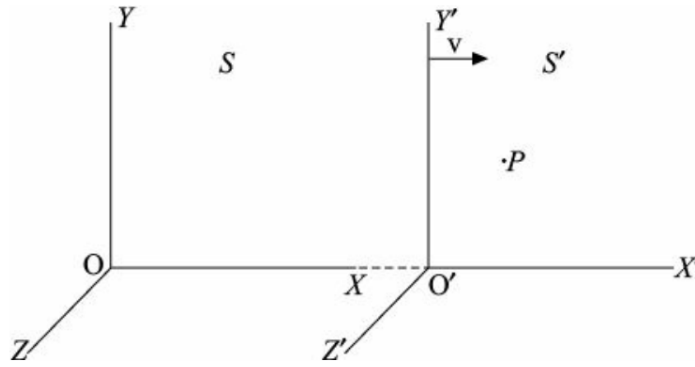
Postulate 1: The principle of equivalence

The laws of physics have the same form in all inertial frames of reference.

Postulate 2: The constancy of the speed of light

The speed of light in free space is always a constant c and is independent of the speed of the source, the observer, or the relative motion of the inertial systems.

3 Lorentz transformation



Consider two inertial frames S and S' with coordinate axes $x y z$ and $x' y' z'$ attached to them, with respective origins O and O' respectively. Let the frame S' be moving with respect to S with a uniform velocity v along the $x x'$ axes as shown in the figure above. The origins of the two frames are assumed to coincide when $t = t' = 0$.

Because the velocity of S' with respect to S is along the x -axis, we have

$$y' = y \quad \text{and} \quad z' = z$$

According to postulate 1, if the motion of a particle is uniformly rectilinear (i.e. in straight line) in frame S , then it must be uniformly rectilinear in frame S' as well. Hence the transformation relating x and x' must be linear, thereby written as

$$x' = \gamma(x - vt) \tag{1}$$

and the inverse transformation is then

$$x = \gamma(x' + vt') \tag{2}$$

Substituting for x' in equation (2) using equation (1), we have

$$\begin{aligned} x &= \gamma[\gamma(x - vt) + vt'] \\ &= \gamma^2 x - \gamma^2 vt + \gamma vt' \\ \Rightarrow \gamma vt' &= \gamma^2 vt + x - \gamma^2 x \\ &= \gamma^2 vt + (1 - \gamma^2)x \\ \Rightarrow t' &= \gamma t + \frac{(1 - \gamma^2)x}{\gamma v} \end{aligned}$$

Then we can write the tentative transformation equations as

$$\boxed{\begin{aligned} x' &= \gamma(x - vt) \\ y' &= y \\ z' &= z \\ t' &= \gamma t + \frac{(1 - \gamma^2)x}{\gamma v} \end{aligned}} \tag{3}$$

Let us now consider an EM wave emitted by a point source located at the position O at time $t = 0$. The spherical wavefront expands with a speed c . Hence the equation of the wavefront in the frame S is

$$x^2 + y^2 + z^2 = c^2 t^2 \quad (4)$$

The corresponding equation in the frame S' is

$$x'^2 + y'^2 + z'^2 = c^2 t'^2 \quad (5)$$

Now, using equations (3) in equation (5), we have

$$\begin{aligned} [\gamma(x - vt)]^2 + y^2 + z^2 &= c^2 \left[\gamma t + \frac{(1 - \gamma^2)x}{\gamma v} \right]^2 \\ \Rightarrow \gamma^2(x - vt)^2 + y^2 + z^2 &= c^2 \left[\gamma^2 t^2 + \frac{(1 - \gamma^2)^2 x^2}{\gamma^2 v^2} + 2\gamma t \frac{(1 - \gamma^2)x}{\gamma v} \right] \\ \Rightarrow \gamma^2 x^2 + \gamma^2 v^2 t^2 - 2\gamma x v t + y^2 + z^2 &= \gamma^2 c^2 t^2 + \frac{(1 - \gamma^2)^2 c^2 x^2}{\gamma^2 v^2} + 2\gamma c^2 t \frac{(1 - \gamma^2)x}{\gamma v} \\ \Rightarrow \gamma^2 x^2 - \frac{(1 - \gamma^2)^2 c^2 x^2}{\gamma^2 v^2} + y^2 + z^2 & \end{aligned}$$